



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI BHAGAVATHI AMMAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

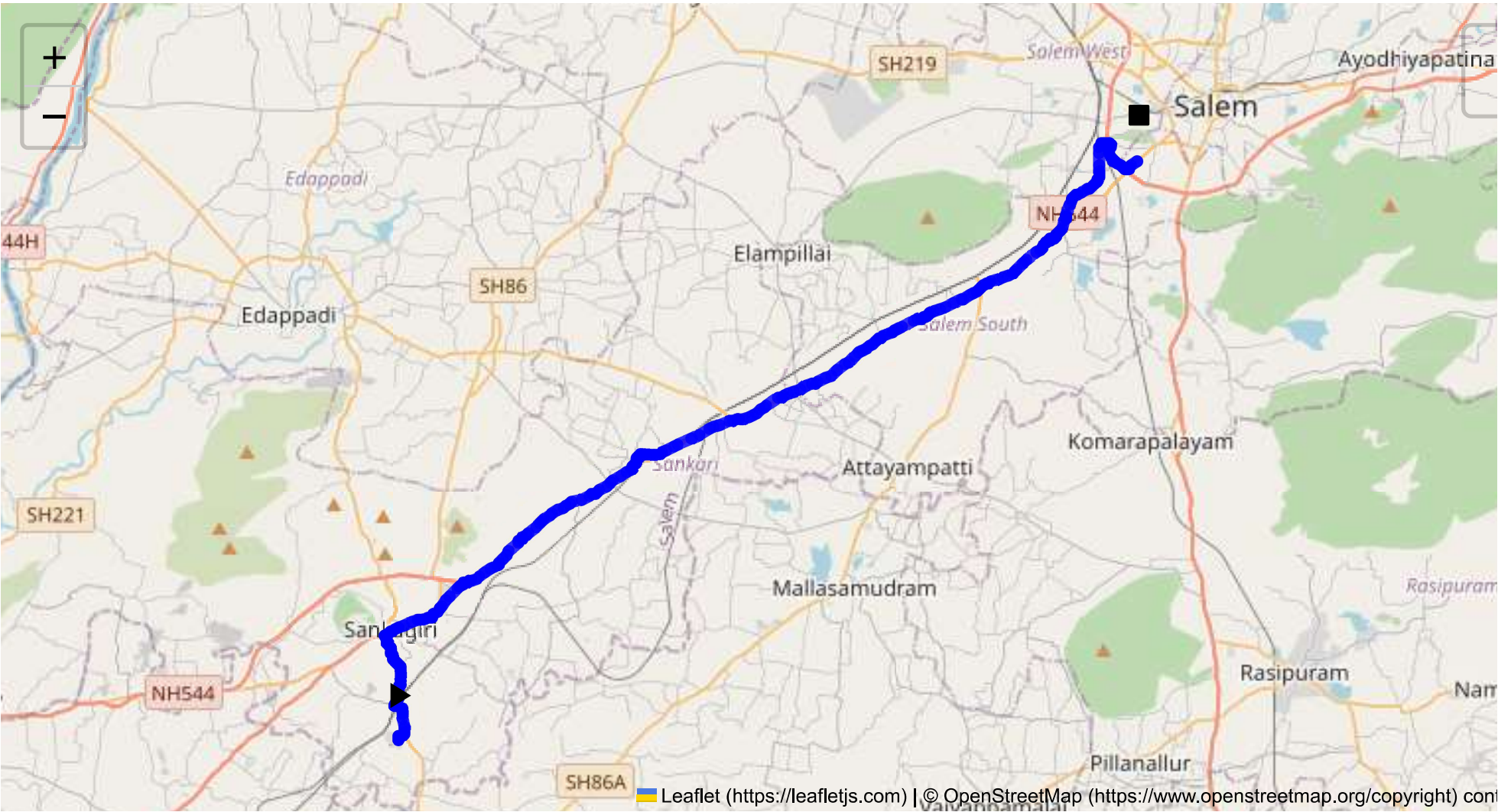
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 41.15 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.636436092900967, 78.13058808774076)



Welcome to the Journey Risk Management Study

Route Overview

The journey from CVQF+23W, Sangagiri, Tamil Nadu to 140, Sankari Main Rd, Salem, Tamil Nadu spans approximately 41.15 kilometers. It primarily follows NH544, a key arterial route known for accommodating substantial commercial traffic, including heavy vehicles.

Weather Conditions and Hazards

- **Typical Weather:** The region generally experiences a tropical climate with the hottest months being March to May. Monsoon season (June to September) brings significant rainfall, potentially leading to road slickness and reduced visibility.
- **Weather-Related Hazards:** During monsoon, the risk of water logging and landslides in some of the hilly stretches could affect the route. Additionally, heavy fog might occur in winter months, impacting visibility.

Traffic Patterns and Congestion

- **Peak Hours:** The route experiences heavy traffic from 8-10 AM and 5-8 PM.
- **Congestion-Prone Areas:** Key congestion points include town entry and exit points and toll booths. The stretch near Salem is particularly congested due to urban density.

Road Quality and Infrastructure

- **Road Quality:** The NH544 is generally well-maintained but local diversions can be problematic.
- **Infrastructure:** Widely available truck stops and service stations but limited emergency pull-offs. Some sections may have construction or repair work affecting traffic flow.

Alternative Routes

In emergencies, consider:

- Bypassing Sangagiri by taking minor roads to rejoin NH544 beyond congestion points.
- Using parallel local roads connecting to Salem if NH544 is blocked or under repair.

Local Regulations for Hazardous Material Transport

- **Regulations:** Regulations demand transport permits for hazardous materials with specifics on containment and required safety measures, including signage and special lighting.
- **Restrictions:** Some routes through densely populated areas may have timings or outright restrictions for hazardous cargo.

Historical Incidents

- **Incident Overview:** Historically, the area along NH544 has recorded a few incidents involving heavy trucks, mostly related to overturning due to sudden braking or maneuvering.
- **Hazardous Material Incidents:** Limited data, but mandatory training for hazardous material handling has minimized incidents.

Environmental Considerations

- **Sensitive Areas:** Proximity to certain ecological and water bodies mandates cautious driving, especially in spill-prone areas.

- **Impact:** Limited impact from standard truck transit but extra care needed with potential hazardous leaks.

Communication Coverage

- **Coverage Analysis:** Most of the route has good mobile network coverage. However, some hilly sections may experience brief dead zones.

Emergency Response Times

- **Response Times:** Emergency services can reach most parts of the route within 30-60 minutes, though rural sections could take longer.

Overall Risk Assessment

The route, while generally secure for heavy vehicles, does have risks related to congestion, weather, and road conditions. Comprehensive preparation, adherence to regulatory compliance, and awareness of environmental considerations are critical in mitigating hazards. Monitoring weather forecasts and traffic updates will aid in safe and timely transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	Medium	11.47410, 77.86845	30 KM/Hr
7	Turn	Medium	11.48366, 77.88781	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.60942, 78.10056	30 KM/Hr
11	Turn	Medium	11.61736, 78.10588	30 KM/Hr
12	Turn	High	11.63314, 78.12745	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
24	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
25	clinic	Salem Care Foundation	11.6371946, 78.1313713	30 km/h	Medium